

1. Discrete Uniform Distribution:

Defⁿ: A discrete r.v. X is said to have a discrete uniform prob distribution over the range $[1, n]$, if its p.m.f. is expressed as follows:

$$P(X=x) = p(x) = \frac{1}{n}, \quad x=1, 2, \dots, n$$

$$= 0, \quad \text{otherwise}$$

Here n is known as the parameter of the distribution and lies in the set of all positive integers. This distribution is also called a discrete rectangular distribution.

Situation where this distⁿ is applicable: If under the given experimental conditions, the different values of the random variable become equally likely. Thus for a die experiment, and for an experiment with a deck of cards such distⁿ is appropriate.

Moments: $A_1' = E(X) = \sum_{x=1}^n x p(x) = \frac{1}{n} \sum_{x=1}^n x = \frac{1}{n} \frac{n(n+1)}{2} = \frac{(n+1)}{2}$

$$A_2' = E(X^2) = \sum_{x=1}^n x^2 p(x) = \frac{1}{n} \sum_{x=1}^n x^2 = \frac{1}{n} \frac{n(n+1)(2n+1)}{6}$$

$$= \frac{(n+1)(2n+1)}{6}$$

$$V(X) = E(X^2) - (E(X))^2 = \frac{(n+1)(n-1)}{12}$$

The MGF of X is $M_X(t) = E e^{tx} = \sum_{x=1}^n e^{tx} p(x)$

$$= \frac{1}{n} \sum_{x=1}^n e^{tx} = \frac{e^t}{n} \left[\frac{(1-e^{nt})}{(1-e^t)} \right]$$

Negative Binomial Distribution (NBD):

In $B(n, p)$, Mean > Variance

In NBD

Mean < Variance

In $P(\lambda)$, Mean = Variance

Application of NBD: Ex. Bacterial clustering & contagion

e.g. deaths of insects, number of insects bites leads to the NBD.

Suppose we have a succession (series) of n Bernoulli trials. We assume that (i) the trials are independent, (ii) the probability of success ' p ' in a trial remains constant from trial to trial.

Let $f(x; r, p)$ denote the probability that there are x failures preceding the r th success in $x+r$ trials.

Now, the last trial must be a success, whose probability is p . In the remaining $(x+r-1)$ trials we must have $(r-1)$ successes whose prob is given by the binomial prob law by the expression: $\binom{x+r-1}{r-1} p^{r-1} q^x$.

Therefore, by the compound probability theorem,

$f(x; r, p)$ is given by the product of these two probabilities

$$\therefore f(x; r, p) = \binom{x+r-1}{r-1} p^{r-1} q^x \cdot p = \binom{x+r-1}{r-1} p^r q^x$$

$$x=0, 1, 2, \dots$$

Definition: A random variable X is said to follow a negative binomial distribution with parameters r and p if its pmf is given by

$$P(X=x) = p(x) = \begin{cases} \binom{x+r-1}{r-1} p^r q^x & ; x=0, 1, 2, \dots \\ 0, & \text{otherwise.} \end{cases} \quad \text{--- (7)}$$

$$\text{Also } \binom{x+r-1}{r-1} = \binom{x+r-1}{x} \quad \left[\binom{n}{r} = \binom{n}{n-r} \right]$$

$$= \frac{(x+r-1)!}{x! (r-1)!} = \frac{(x+r-1)(x+r-2)\dots(r+1)r \cancel{(r-1)!}}{x! \cancel{(r-1)!}}$$

$$= \frac{(-1)^x (-r)(-r-1)\dots(-r-x+2)(-r-x+1)}{x!} = (-1)^x \binom{-r}{x}$$

$$\therefore p(x) = \begin{cases} \binom{-r}{x} p^r (-q)^x & ; x=0, 1, 2, \dots \\ 0 & \text{otherwise} \end{cases}$$

which is $(x+1)^{\text{th}}$ term in the expansion of $p^r (1-q)^{-r}$, a binomial expansion with a negative index. Hence the distribution is known as negative binomial distⁿ.

$$\text{Also } \sum_{x=0}^{\infty} p(x) = p^r \sum_{x=0}^{\infty} \binom{-r}{x} (-q)^x = p^r (1-q)^{-r} = 1.$$

Hence $p(x)$ is pmf.

If $p = \frac{1}{Q}$ and $q = \frac{P}{Q}$, so that $Q - P = 1$, ($\because p + q = 1$), then

$$p(x) = \begin{cases} \binom{-r}{x} Q^{-r} \left(-\frac{P}{Q}\right)^x; & x = 0, 1, 2, \dots \\ 0, & \text{otherwise} \end{cases}$$

This is the general term in the negative binomial expansion $(Q - P)^{-r}$.

Remark: (1). If we take $r=1$ in $(*)$, we have

$$p(x) = q^x p; \quad x = 0, 1, 2, \dots$$

which is the prob function of geometric distⁿ. Hence NBD may be regarded as the generalization of geometric distⁿ.

Moment Generating Function of Negative Binomial Distribution.

$$M_X(t) = E e^{tx} = \sum_{x=0}^{\infty} e^{tx} p(x) = \sum_{x=0}^{\infty} \binom{-r}{x} Q^{-r} \left(-\frac{Pe^t}{Q}\right)^x \\ = (Q - Pe^t)^{-r}$$

$$\text{Mean } h_1' = \frac{d}{dt} M_X(t) \Big|_{t=0} = \left[-r(-Pe^t)(Q - Pe^t)^{-r-1} \right]_{t=0} = rP$$

$$\underline{\text{Mean} = rP}$$

$$h_2' = \frac{d^2 M_X(t)}{dt^2} \Big|_{t=0} = \left[rPe^t(Q - Pe^t)^{-r-1} + (-r-1)rPe^t(Q - Pe^t)^{-r-2}(-Pe^t) \right]_{t=0} \\ = rP + r(r+1)P^2$$

$$\underline{\text{Variance}} \quad \therefore h_2 = h_2' - h_1'^2 = rP + r(r+1)P^2 - r^2P^2 = rPQ$$

As $Q > 1$, $rP < rPQ$ i.e. Mean < Variance

Which is distinguishing feature of the NBD.

Ex: An item is produced in large numbers. The machine is known to produce 5% defective. A quality control inspector is examining the items by taking them at random. What is the probability that at least 4 items are to be examined in order to get 2 defectives?

Solⁿ: If 2 defectives are to be obtained then it can happen in 2 or more trials. The prob. of success is 0.05 for every trial. It is negative binomial ~~dist~~ situation with $r=2$, $p=0.05$ and x replaced by $(x-2)$ and the required prob is given by

$$P(X=4) + P(X=5) + \dots = \sum_{x=4}^{\infty} \binom{x-1}{2-1} (0.05)^2 (0.95)^{x-2}$$

$$= 1 - \sum_{x=2}^3 \binom{x-1}{2-1} (0.05)^2 (0.95)^{x-2} = 0.9928$$

$$X+Y \geq 4$$

$$X+Y = U$$

$$X = U - Y = U - 2$$

$$P(U \geq 4) = \sum_{u=4}^{\infty} \binom{u-1}{2-1} (0.05)^2 (0.95)^u$$

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Geometric Distribution:

Suppose we have a series of independent bernoulli trials or repetitions and in each trial the prob. of success ' p ' remains the same. Then the probability that there are x failures preceding the first success is given by $q^x p$, $q=1-p$.

Defⁿ: A random variable X is said to have a geometric distribution if it assumes only non-negative values and its prob mass function is given by

$$P(X=x) = p(x) = \begin{cases} q^x p; & x=0, 1, 2, \dots \\ 0, & \text{otherwise} \end{cases}$$

$$0 < p \leq 1; \quad q = 1-p.$$

Remark: (i) The various probabilities for $x=0, 1, 2, \dots$ are the terms of geometric series, hence the name geometric distⁿ.

(ii) $\sum_{x=0}^{\infty} p(x) = \sum_{x=0}^{\infty} q^x p = p(1+q+q^2+\dots) = \frac{p}{1-q} = \frac{p}{p} = 1$, hence it's pmf.

Lack of Memory: The geometric distribution is said to lack memory in a certain sense. Suppose an event E can occur at one of the times $t = 0, 1, 2, \dots$ and the occurrence (waiting) time X has a geometric distribution with parameter p

$$P(X=t) = q^t p; t=0, 1, 2, \dots$$

Suppose we know that the event E has not occurred before K , i.e., $X \geq K$. Let $Y = X - K$. Thus, Y is the amount of additional time needed for E to occur. We can show that

$$P(Y=t | X \geq K) = P(X=t) = p q^t$$

which implies that the additional time to wait has the same distⁿ as initial time to wait.

Since the distribution does not depend K , it's, in a sense, 'lack memory' of how much we shifted the time origin. If 'B' were waiting for the event E and is relieved by 'C' immediately before time K , then the waiting time distribution of 'C' is the same as that of 'B'.

Proof:
$$P(X \geq r) = \sum_{k=r}^{\infty} p q^k = p (q^r + q^{r+1} + q^{r+2} + \dots) = \frac{p q^r}{(1-q)} = q^r \quad \text{--- (x)}$$

$$\begin{aligned} P(Y \geq t | X \geq K) &= \frac{P(Y \geq t \cap X \geq K)}{P(X \geq K)} = \frac{P(X-K \geq t \cap X \geq K)}{P(X \geq K)} \\ &= \frac{P(X \geq K+t)}{P(X \geq K)} = \frac{q^{K+t}}{q^K} = q^t \quad \text{--- (y)} \end{aligned}$$

($\because Y = X - K$)
(From (x))

$$\begin{aligned} \therefore P(Y=t | X \geq K) &= P(Y \geq t | X \geq K) - P(Y \geq t+1 | X \geq K) \\ &= q^t - q^{t+1} = q^t (1-q) = p q^t = P(X=t) \end{aligned}$$

(From (y))

Moment Generating Function of Geometric Distribution

$$M_X(t) = E e^{tX} = \sum_{x=0}^{\infty} e^{tx} q^x p = p \sum_{x=0}^{\infty} (qe^t)^x = p(1 - qe^t)^{-1}$$

$$= \frac{p}{1 - qe^t}$$

$$\text{Mean } \mu_1' = \left. \frac{d}{dt} M_X(t) \right|_{t=0} = \left[\frac{d}{dt} p(1 - qe^t)^{-1} \right]_{t=0} = pq(1 - q)^{-2} = \frac{q}{p}$$

$$\mu_2' = \left. \frac{d^2}{dt^2} M_X(t) \right|_{t=0} = \frac{q}{p} + \frac{2q^2}{p^2} \quad (\text{after simplification})$$

$$\therefore \text{Variance } \mu_2 = \mu_2' - \mu_1'^2 = \frac{q}{p} + \frac{2q^2}{p^2} - \frac{q^2}{p^2} = \frac{q}{p} + \frac{q^2}{p^2} = \frac{pq + q^2}{p^2}$$

$$= \frac{q}{p^2}$$

Hence the mean and variance of geometric distribution are $\frac{q}{p}$ and $\frac{q}{p^2}$ respectively.

Remark: Variance = $\frac{q}{p^2} = \frac{q}{p} \cdot \frac{1}{p} = \frac{\text{Mean}}{p} > \text{Mean}$

Hence, in geometric distⁿ Variance > Mean.

Ex. Let X_1, X_2 be independent r.v.s each having geometric distribution $q^k p$; $k=0, 1, 2, \dots$ Show that the conditional distribution of X_1 given $(X_1 + X_2)$ is Uniform.

$$\text{Sol}^n. P(X_1 = r | X_1 + X_2 = n) = \frac{P(X_1 = r \cap X_1 + X_2 = n)}{P(X_1 + X_2 = n)}$$

$$= \frac{P(X_1 = r \cap X_2 = n - r)}{P(X_1 + X_2 = n)} = \frac{P(X_1 = r \cap X_2 = n - r)}{\sum_{\lambda=0}^n P(X_1 = \lambda \cap X_2 = n - \lambda)}$$

$$= \frac{P(X_1 = r) P(X_2 = n - r)}{\sum_{\lambda=0}^n [P(X_1 = \lambda) P(X_2 = n - \lambda)]}$$

Since X_1 and X_2 are independent.

$$\therefore P[X_1 = r | (X_1 + X_2 = n)] = \frac{pq^r \cdot pq^{n-r}}{\sum_{\lambda=0}^n [pq^\lambda \cdot pq^{n-\lambda}]} = \frac{p^2 q^n}{\sum_{\lambda=0}^n p^2 q^n} = \frac{p^2 q^n}{(n+1)p^2 q^n} = \frac{1}{n+1}; r=0, 1, \dots, n.$$

Hence, the result.

Ex Suppose X is a non-negative integral random variable. Show that the distribution of X is geometric if it 'lacks memory' i.e. if for each $k \geq 0$ and $Y = X - k$, one has $P(Y=t | X \geq k) = P(X=t)$, for $t \geq 0$.

Solⁿ. Let us suppose $P(X=x) = p_x$; $x = 0, 1, 2, \dots$

Define $q_k = P(X \geq k) = p_k + p_{k+1} + p_{k+2} + \dots$

We are given $P(Y=t | X \geq k) = P(X=t) = p_t$

$$\begin{aligned} P(Y=t | X \geq k) &= \frac{P(Y=t \cap X \geq k)}{P(X \geq k)} = \frac{P(X-k=t \cap X \geq k)}{P(X \geq k)} \\ &= \frac{P(X=t+k)}{P(X \geq k)} = \frac{p_{k+t}}{q_k} \end{aligned}$$

$\therefore p_t = \frac{p_{k+t}}{q_k}$, for every $t \geq 0$ and all $k \geq 0$ — From (1)

In particular, taking $k=1$, we get

$$p_{t+1} = q_1 p_t = (p_1 + p_2 + \dots) p_t = (1 - p_0) p_t \quad \text{From (1)}$$

$$\Rightarrow p_t = (1 - p_0) p_{t-1} = (1 - p_0)^2 p_{t-2} = \dots = (1 - p_0)^t p_0$$

$$\text{Hence } p_t = P(X=t) = p_0 (1 - p_0)^t; \quad t = 0, 1, 2, \dots$$

$\Rightarrow X$ has a geometric distribution.

Hyper. Geometric Distribution:

Consider an urn with N balls, M of which are white and $N-M$ are red. Suppose that we draw a sample of n balls at random (without replacement) from the urn. Then the prob of getting K white balls out of n , ($K \leq n$) is

$$\frac{\binom{M}{K} \binom{N-M}{n-K}}{\binom{N}{n}}$$

[Since K white balls can be drawn from ' M ' white balls in $\binom{M}{K}$ ways and out of the remaining $N-M$ red balls, $(n-K)$ balls can be chosen in $\binom{N-M}{n-K}$ ways, the total number of favourable cases is $\binom{M}{K} \binom{N-M}{n-K}$]

Defⁿ: A discrete r.v X is said to follow the hypergeometric distribution with parameters N, M and n if it assumes only non-negative values and its prob mass function is given by:

$$P(X=k) = p(k; N, M, n) = \begin{cases} \frac{\binom{M}{k} \binom{N-M}{n-k}}{\binom{N}{n}} & k=0, 1, \dots, \min(n, M) \\ 0 & \text{otherwise} \end{cases}$$

Where N is a positive integer, M is a positive integer not exceeding N and n is a positive integer that is at most N .

As it can be shown that $\sum_{k=0}^n \binom{M}{k} \binom{N-M}{n-k} / \binom{N}{n} = 1$.

Mean $E(X) = \sum_{k=0}^n k P(X=k) = \sum_{k=0}^n k \left\{ \frac{\binom{M}{k} \binom{N-M}{n-k}}{\binom{N}{n}} \right\}$

$$= \frac{nM}{N}$$

$$E(X^2) = \sum_{k=0}^n k^2 P(X=k) = \sum_{k=0}^n \{k(k-1) + k\} P(X=k)$$

$$= \frac{M(M-1)n(n-1)}{N(N-1)} \quad (\text{on simplification})$$

$$\therefore V(X) = E(X^2) - (E(X))^2 = \frac{NM(N-M)(N-n)}{N^2(N-1)} \quad (\text{on simplification})$$